A SHARPER CERTIFIED LOWER BOUND FOR ERDŐS' MINIMUM-OVERLAP PROBLEM

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Abstract. We prove the computer-assisted lower bound

$$c_E > 0.38056070$$

for Erdős' minimum-overlap constant. The proof uses the 172-bin mean-conditioned certificate framework of Price, retaining his certified Arb bounds on the 170 noncentral bins and replacing the two binding center bins. The new center certificate strengthens the earlier Parseval hybrid by replacing a hard spectral cutoff with a fixed weighted Parseval row

$$-\sum_{n=1}^{400} w_n C_{n\pi} \le \frac{1}{2}, \qquad 0 \le w_n \le 1,$$

and by combining the weighted-Parseval and exact harmonic rows into 400 finite-decimal coefficients c_n. The frozen proof object contains one second-moment multiplier, 80 ordinary cosine multipliers, one positive weighted-Parseval budget λ_W, and 400 coefficients satisfying $c_n \le \lambda_W$. This inequality is sufficient for a valid decomposition $c_n = \lambda_W w_n - \eta_n$ with $0 \le w_n \le 1$ and $\eta_n \ge 0$. A direct MPFR verification gives

$$\int_{-2}^{2} q_+(t)\,dt \le 2.62770156549654007831192522588 3\ldots < \frac{1}{0.38056070}.$$

The optimizer used to discover the certificate is outside the trusted proof path.

1. Continuous formulation

For a partition $A \sqcup B = \{1, \ldots, 2N\}$ with $|A| = |B| = N$, put

$$r_{A,B}(m) = |\{(a, b) \in A \times B : a - b = m\}|, \qquad R(A, B) = \max_m r_{A,B}(m),$$

and

$$c_E = \liminf_{N\to\infty} \frac{1}{N} \min_{A\sqcup B=\{1,\ldots,2N\}} R(A, B).$$

Following White and Price [1, 2], let $\mathcal{H}$ be the measurable functions $h : [0, 2] \to [0, 1]$ with $\int_0^2 h = 1$, put $g = 1 - h$ on $[0, 2]$, extend both by zero, and define

$$p_h(t) = \int_{\mathbb{R}} h(x)g(x + t)\,dx, \qquad -2 \le t \le 2.$$

Then $p_h \ge 0$, $\int p_h = 1$, and $\operatorname{supp} p_h \subset [-2, 2]$. Write

$$c_{\text{cont}} = \inf_{h\in\mathcal{H}} \|p_h\|_\infty.$$

Lemma 1 (Discrete-to-continuous reduction)**.** $c_E \ge c_{\text{cont}}$.

Proof. For $I_k = [(k-1)/N, k/N)$ set $h_N = \sum_{a\in A} \mathbf{1}_{I_a}$ and $g_N = \sum_{b\in B} \mathbf{1}_{I_b}$. With $\tau(u) = (1-|u|)_+$,

$$p_{h_N}(t) = \frac{1}{N} \sum_{a\in A,b\in B} \tau(Nt - (b - a)).$$

For fixed x, the nonzero values among $\tau(x - d)$, $d \in \mathbb{Z}$, sum to at most one. Grouping by $d = b - a$ gives $\|p_{h_N}\|_\infty \le R(A, B)/N$. Minimize and take the liminf. □

Date: 7 September 2026.

Fix $h \in \mathcal{H}$, write $p = p_h$, and let

$$\mu = \int_{-2}^{2} tp(t)\, dt.$$

The elementary rearrangement bound $1/2 \le \mathbb{E}X \le 3/2$ for a random variable with density h implies $\mu \in [-1, 1]$.

2. Analytic rows

2.1. Second moment.

Lemma 2. *For every admissible p,*

$$\int t^2 p(t)\, dt = \frac{2}{3} + \frac{\mu^2}{2}.$$

Hence on $|\mu| \le 1/320$,

$$\int t^2 p(t)\, dt \le B_2 := \frac{2}{3} + \frac{1}{2 \cdot 320^2}.$$

Proof. Let X, Y be independent with densities h, g. Then p is the density of $Y - X$, while $\mathbb{E}X + \mathbb{E}Y = 2$ and $\mathbb{E}X^2 + \mathbb{E}Y^2 = 8/3$. Since $\mathbb{E}Y - \mathbb{E}X = \mu$, one has $\mathbb{E}X = 1 - \mu/2$, $\mathbb{E}Y = 1 + \mu/2$, and the displayed identity follows. □

2.2. Fourier rows.

Define

$$H(\xi) = \int_0^2 h(x) e^{i\xi(x-1)}\, dx, \qquad G(\xi) = \int_0^2 g(x) e^{i\xi(x-1)}\, dx,$$

and

$$C_\xi = \int_{-2}^{2} p(t) \cos(\xi t)\, dt.$$

Since $H(\xi) + G(\xi) = 2 \operatorname{sinc} \xi$ and $\int p(t) e^{i\xi t}\, dt = \overline{H(\xi)} G(\xi)$, the usual completion of squares gives the following.

Lemma 3 (Pointwise cosine bound)**.** *For every real ξ,*

$$C_\xi \le \operatorname{sinc}^2 \xi.$$

In particular, for every positive integer n,

$$C_{n\pi} = -|H(n\pi)|^2 \le 0.$$

2.3. Weighted Parseval.

The next form of White's Parseval restriction is the only new ingredient in the certificate language relative to the previous hard-cutoff version.

Lemma 4 (Weighted Parseval row)**.** *Let (w_n) be finitely supported with $0 \le w_n \le 1$. Then*

$$-\sum_{n \ge 1} w_n C_{n\pi} \le \frac{1}{2}.$$

Proof. At $n\pi$, Lemma 3 gives $-C_{n\pi} = |H(n\pi)|^2$. Parseval for $u \mapsto h(u+1)$ on $[-1, 1]$ gives

$$1 + 2 \sum_{n \ge 1} |H(n\pi)|^2 \le 2,$$

because $H(0) = 1$ and $0 \le h \le 1$ implies $\int h^2 \le \int h = 1$. Therefore

$$-\sum_n w_n C_{n\pi} = \sum_n w_n |H(n\pi)|^2 \le \sum_n |H(n\pi)|^2 \le \frac{1}{2}.$$

□

Remark 5. *The underlying Parseval energy inequality is due to the existing Fourier framework and is not claimed as new. The present improvement is the use of a frozen non-rectangular spectral window inside Price's mean-conditioned dual certificate.*

3. Dual certificate

Lemma 6 (Positive-part dual bound)**.** *Suppose p satisfies finitely many valid inequalities $\int p\phi_j \le B_j$. For $\lambda_j \ge 0$ put*

$$q(t) = 1 + \sum_j \lambda_j (B_j - \phi_j(t)), \qquad q_+(t) = \max(q(t), 0).$$

Then

$$\|p\|_\infty \ge \frac{1}{\int_{-2}^{2} q_+(t)\,dt}.$$

Proof. $\int pq \ge 1$, while $p \ge 0$ implies $\int pq \le \|p\|_\infty \int q_+$. □

4. The frozen center certificate

Price's 172-bin cover has the binding pair

$$I_- = [-1/320, 0], \qquad I_+ = [0, 1/320].$$

The authoritative machine-readable proof object is `certificate/weighted_center_combined_038056070.txt`. It contains exact finite-decimal strings for

- one nonnegative second-moment multiplier λ_2;
- 80 nonnegative ordinary cosine multipliers λ_ξ;
- one positive weighted-Parseval budget λ_W;
- 400 combined coefficients $c_1, \ldots, c_{400}$ satisfying $c_n \le \lambda_W$.

The associated even trigonometric polynomial is

$$q(t) = 1 + \lambda_2(B_2 - t^2) + \sum_{\xi \in \Xi} \lambda_\xi (\operatorname{sinc}^2 \xi - \cos(\xi t)) + \frac{\lambda_W}{2} + \sum_{n=1}^{400} c_n \cos(n\pi t).$$

The combined form is itself a valid proof object. Indeed, for every n define

$$(w_n, \eta_n) = \begin{cases} (c_n/\lambda_W, 0), & c_n \ge 0, \\ (0, -c_n), & c_n < 0. \end{cases}$$

Then $0 \le w_n \le 1$, $\eta_n \ge 0$, and $c_n = \lambda_W w_n - \eta_n$. Consequently the last line of q is exactly the sum of a valid weighted-Parseval row and nonnegative multiples of the exact harmonic rows $C_{n\pi} \le 0$. The script `code/check_weighted_combined_certificate.py` verifies the finite-decimal sign conditions, the complete index set $1, \ldots, 400$, and $c_n \le \lambda_W$ using exact rational arithmetic. The optimizer used to find the coefficients is not needed for this check.

5. Certified direct-MPFR verification

`code/verify_weighted_center_mpfr.c` uses MPFR with explicit directed rounding. At a cell center it encloses $q, q', \ldots, q^{(6)}$ and uses a global seventh-derivative bound for the Taylor remainder. Cells proved positive are integrated with the explicit elementary antiderivative of q; terminal ambiguous cells are charged conservatively by their width times a rigorous upper bound for q. The 32 subinterval endpoints in `verification/weighted_038056070_center_manifest.csv` cover $[0, 2]$ exactly. The runner recomputes every chunk from the frozen combined certificate, parses each printed upward-rounded bound as an exact rational, and then sums and doubles by evenness. A clean reproduction gives

$$\int_{-2}^{2} q_+(t)\,dt \le 2.62770156549654007831192522588 3\ldots < \frac{1}{0.38056070}. \tag{1}$$

More precisely, the D-side margin against $1/0.38056070$ is greater than 1.38857×10^{-7}. Thus every admissible p with $\mu \in I_- \cup I_+$ satisfies

$$\|p\|_\infty > 0.38056070.$$

6. Noncentral bins and the theorem

For the other 170 mean bins we reuse Price's published Arb-certified upper bounds [2]. The largest such value is attained at bins 77 and 94:

$$D_{\text{noncentral}} \le 2.62753853087337579009027217587 84\ldots,$$

so

$$\frac{1}{D_{\text{noncentral}}} \ge 0.3805843333028524\ldots > 0.38056070.$$

The release script `code/check_noncentral_target.py` parses all 170 vendored Arb balls and checks this comparison exactly from their printed midpoint-radius enclosures.

Theorem 7. *The Erdős minimum-overlap constant satisfies*

$$\boxed{c_E > 0.38056070}.$$

Proof. Every mean $\mu \in [-1, 1]$ belongs to Price's 172-bin cover. On the two center bins, (1) and Lemma 6 give the stated bound. On every other bin the reused Arb certificate gives the stronger noncentral bound above. Hence $c_{\text{cont}} > 0.38056070$, and Lemma 1 implies the theorem. □

7. Reproducibility and trust model

The trusted finite data are the standalone combined center certificate, the vendored Price Arb reports for the 170 noncentral bins, and the source code of the exact certificate checker and direct-MPFR verifier. The LP search, floating-point continuous integration, and historical Python interval experiments are exploratory only.

The release provides:

- exact rational validation of the combined certificate, including $c_n \le \lambda_W$ for all $1 \le n \le 400$;
- a direct-MPFR verifier and exact chunk aggregator;
- a target check for all 170 noncentral Arb reports;
- SHA-256 hashes for the release files and a reproducible PDF build.

The center verifier was developed within the same project; this is not a third-party audit. External reproduction remains desirable.

Use of AI.. Computational exploration, certificate search, verifier engineering, and parts of the exposition were carried out with substantial assistance from OpenAI's ChatGPT. The language model is not part of the trusted proof path. The human author is responsible for the mathematical claims and release.

References

[1] E. P. White, *A new bound for Erdős' minimum overlap problem*, Acta Arith. **208** (2023), 235–255. doi:10.4064/aa220728-7-6.

[2] L. Price, *A Certified Lower Bound for Minimum Overlap*, proof note and reproducible Arb certificate for $c_E > 0.38055470$, 29 June 2026. `https://github.com/Leeham06972452/erdos-36-lower-bound`.

[3] occisn, *Independent verification of the* 0.38055470 *minimum-overlap certificate*, 2026. `https://github.com/occisn/erdos-36-certified-lower-bound`.

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